

Sensitized by a sea slug: Site-specific short-term and general long-term sensitization in *Aplysia* following *Navanax* attack

Chrissy Pepino, Cyril Rakovski, Candace Gutierrez, Amanda Rodriguez, Scott Tillett, Cassandra Berriman, Maria Mason, Amanda Watkins Ingalls, Rebecca Emshwiler, Sarah Christian Scher, Virginia Zachary, Lucy Lee, Sarah Johnson, Kendra Olsen, William G. Wright^{*}

Schmid College of Science and Technology, Chapman University, United States

ARTICLE INFO

Keywords:

Learning and memory
Evolution
Predation
Long-term memory
Short-term memory
Attack
Noxious stimulus

ABSTRACT

Neurobiological studies of the model species, *Aplysia californica* (Mollusca, Gastropoda, Euopisthobranchia), have helped advance our knowledge of the neural bases of different forms of learning, including sensitization, a non-associative increase in withdrawal behaviors in response to mild innocuous stimuli. However, our understanding of the natural context for this learning has lagged behind the mechanistic studies. Previous studies, which exclusively used artificial stimuli, such as electric shock, to produce sensitization, left open the question of which stimuli might cause sensitization in nature. Our laboratory first addressed this question by testing for short and long-term sensitization after predatory attack by a natural predator, the spiny lobster. In the present study, we tested for sensitization after attack by a very different predator, the predacious sea-slug, *Navanax inermis* (Mollusca, Gastropoda, Euopisthobranchia).

Unlike the biting and prodding action of lobster attack, *Navanax* uses a rapid strike that sucks and squeezes its prey in an attempt to swallow it whole. We found that *Navanax* attack to the head of *Aplysia* caused strong immediate sensitization of head withdrawal, and weaker, delayed, sensitization of tail-mantle withdrawal. By contrast, attack to the tail of *Aplysia* resulted in no sensitization of either reflex. We also developed an artificial attack stimulus that allowed us to mimic a more consistently strong attack. This artificial attack produced stronger but qualitatively similar sensitization: Strong immediate sensitization of head withdrawal and weaker sensitization of tail-mantle withdrawal after head attack, immediate sensitization in tail-mantle withdrawal, but no sensitization of head withdrawal after tail attack. We conclude that *Navanax* attack causes robust site-specific sensitization (enhanced sensitization near the site of attack), and weaker general sensitization (sensitization of responses to stimuli distal to the attack site).

We also tested for long-term sensitization (lasting longer than 24 h) after temporally-spaced delivery of four natural *Navanax* attacks to the head of subject *Aplysia*. Surprisingly, these head attacks, any one of which strongly sensitizes head withdrawal in the short term, failed to sensitize head-withdrawal in the long term. Paradoxically, these repeated head attacks produced long-term sensitization in tail-mantle withdrawal.

These experiments and observations confirm that *Navanax* attack causes short, and long-term sensitization of withdrawal reflexes of *Aplysia*. They add site-specific sensitization as well as paradoxical long-term sensitization of tail-mantle withdrawal to a short list of naturally induced learning phenotypes in this model species. Together with previous observations of sensitization after lobster attack, these data strongly support the premise that sensitization in *Aplysia* is an adaptive response to sub-lethal predator attack.

1. Introduction

Our understanding of the mechanistic underpinnings of behavioral

change after an intervening experience, i.e., learning, has been well served by a variety of animal model species. One of these species, the gastropod mollusk *Aplysia californica*, has produced an exceedingly rich

^{*} Corresponding author at: Chapman University, One University Drive, Orange, CA 92866, United States.

E-mail address: wwright@chapman.edu (W.G. Wright).

<https://doi.org/10.1016/j.nlm.2021.107542>

Received 27 April 2021; Received in revised form 13 October 2021; Accepted 22 October 2021

Available online 5 November 2021

1074-7427/© 2021 Elsevier Inc. All rights reserved.

literature, especially regarding a simple form of learning, sensitization. Sensitization refers to an increase in the strength of withdrawal responses to innocuous stimuli (e.g., head withdrawal in response to water jet stimuli), after a novel or noxious stimulus (e.g. electric shock). It has been intensively studied in *Aplysia* at behavioral, synaptic, cellular, and molecular levels (Reviews by Antzoulatos & Byrne, 2004; Barco, Bailey, & Kandel, 2006; Kandel, 2001, 2012; Rahn, Guzman-Karlsson, & Sweatt, 2013; Reissner, Shobe, & Carew, 2006; Roberts & Glanzman, 2003).

Typically, sensitization is produced in an individual *Aplysia* by applying a strong electric shock to its body wall. Following such a shock, multiple withdrawal reflexes are enhanced, i.e., sensitized. When the reflex tested is elicited by stimuli anatomically distant from the site of the shock, the observed enhancement after the shock is labeled “general” sensitization (Walters, 1991; Sutton, Bagnall, Sharma, Shobe, & Carew, 2004). When the reflex tested is elicited by stimuli anatomically close to the shock, any increased enhancement which is greater than the enhancement of reflexes initiated far from the shock, is referred to as “site-specific” sensitization (Walters, 1991; Sutton et al., 2004; Marinesco, Duran, & Wright, 2003). Thus, site-specific sensitization is more robust (greater and/or more long-lasting), and appears less than 10 min after the noxious stimulus (Walters, 1987b, E.T. Walters pers. comm. to WG). By contrast, general sensitization is weaker, and its expression is typically delayed for 15–50 min after the noxious stimulus (Mackey et al., 1987; Marcus, Nolen, Rankin, & Carew, 1988; Wright, McCance, Theresa, & Carew, 1992).

Although electric shock is a highly repeatable stimulus that reliably causes both general and site-specific sensitization, it is also artificial, leaving unanswered the question, “What natural stimuli might cause sensitization?” Watkins et al. (2010) first addressed this question by finding short-term sensitization in *Aplysia* after attack by the spiny lobster *Panulirus interruptus*. This sensitization had a delayed onset, similar to that observed in general sensitization after electric shock (Mackey et al., 1987; Marcus et al., 1988; Wright et al., 1992). Furthermore, multiple lobster attacks caused long-term sensitization and neural correlates very much like those after multiple electric shocks.

Spiny lobsters (*Panulirus* spp.) represent just one of myriad known predators on *Aplysia* (e.g., anemones: Nolen, Johnson, Kicklighter, & Capo, 1995, pycnogonid crabs: Rogers, De Nys, & Steinberg, 2000, multiple fish species: Pennings, Nastisch, & Paul, 2001; Nusbaum & Derby, 2010; green crab: Kamio, Grimes, Hutchins, van Dam, & Derby, 2010; *Navanax inermis*: Paine, 1965; Susswein, Achituv, Cappell, Spray, & Bennett, 1984; Pennings, 1990). These predators employ heterogeneous attack and consumption strategies on *Aplysia*, raising the question of whether sensitization displayed by *Aplysia* following attacks by them might also be heterogeneous.

Here, we investigated whether attack on *Aplysia* by a co-occurring (Pennings, 1990, Wright, pers. obs.) sea-slug predator which utilizes an attack strategy far different from that of the spiny lobster (see Results, Paine, 1965; Blair & Seapy, 1972; Murray & Lewis, 1974; Susswein et al., 1984, 1987; Pennings, 1990), can also cause sensitization in *Aplysia*.

2. Materials and methods

We collected specimens of *Navanax inermis* with mask and snorkel (depth 0.5–4.0 m) from Alamitos Bay, Long Beach CA. Specimens of *Aplysia californica* were collected by a professional collector (Alacrity Marine Biological Services, Redondo Beach, CA). Individual *Navanax* were kept isolated for 3–8 d before being used. Both species were kept in artificial seawater (Instant Ocean; Aquarium Systems, Inc., 14–16 °C).

2.1. Testing

First, two pre-test stimuli (0.5 s water-jets pre-adjusted to cause a modest withdrawal reflex) were delivered to subject *Aplysia* with a “water-pik” (Teledyne Inc), alternately to the head and tail (2.5 min

interval) of a pair of subject *Aplysia* (order randomized), and reflexes recorded to video for later playback/scoring. Then, each of the pair was randomly designated as either Attack (exposed to *Navanax* attack, see below) or No Attack (placed in a tank containing *Navanax* for the same amount of time, but no attack delivered). “Post-test” water-pik stimuli were delivered 5, 15, 30, and 60 min after the end of the attack. During video playback of reflex withdrawals, an observer, with no knowledge of the attack status of the subjects, scored the duration of head and tail-mantle withdrawal (Watkins et al., 2010; Mason et al., 2014). For head withdrawal this comprised the elapsed time from stimulus onset until the retracted head, propodia, and rhinophores relaxed to ca 70% of their original position. For tail-mantle withdrawal, this comprised the time until the retracted tail, siphon, mantle, and parapodia relaxed to 70% of their original position (Wright, 1998; Mason et al., 2014).

2.2. Attack

Each experimental *Aplysia* was subjected to *Navanax* attack by one of approximately 20–40 different individual *Navanax*. Because they attacked sea hares, even if held in the hand of the experimenter (Video 4), we were able to maneuver individual *Navanax* so as to specifically attack the head or tail of subject *Aplysia*. We chose *Navanax* individuals (40–90 mm) that were 0.75–1.5 times the length of the subject *Aplysia* (30–130 mm). Individuals that are too small cannot hold attacked *Aplysia* more than a few seconds (see 0:59 of Video 1). Individuals that are too large immediately engulf the entire *Aplysia* (Video 3).

Sixty seconds after initiation of the attack, attacking *Navanax* were carefully removed from subject *Aplysia*. If an attacking *Navanax* released the subject *Aplysia* before 60 s had elapsed, additional attacks (no more than four were ever needed) were immediately initiated until the subject had accumulated 60 s of attack.

In a different set of experiments, we developed a proxy that more consistently mimicked the physical actions of *Navanax* attack (“artificial *Navanax* attack”) by wrapping subject *Aplysia* head or tail regions snugly into a neonatal blood-pressure cuff and increasing the cuff pressure to 70–80 mm Hg for 1 min. For these experiments, we used larger subject *Aplysia* (100–140 mm), such that the pressure cuff width (75 mm) could be fit onto the anterior or posterior end of the subject.

2.3. Long-term sensitization

We pre-tested each subject, as in the short-term sensitization experiments (see above). Then half were given four separate 60-second natural *Navanax* attacks to the head; the other half were treated as handling controls. Because we sometimes had a limited number of *Navanax*, and because each *Navanax* was only capable of a limited number of attacks in a 1–2 h time frame, we chose to distribute attacks across 2 days (cf. Mason et al., 2014). This meant one to three attacks on the first day and one to three attacks on the second day, for a total of four attacks. Attacks on the same day were separated by 1–2 h; the first attack on the second day was 13–21 h after the last attack on the first day. Post-tests were delivered 24 h after the last attack.

We quantified contraction (“ball-up”) of subject *Aplysia* to each head-attack (during long-term training) as the length of time elapsed from the end of each attack until the subject *Aplysia* had relaxed to ca 70% of its initial appearance before the attack.

2.4. Analysis

We used R (R-core team 2020) for all statistical tests.

2.4.1. Short-term sensitization

The post-attack increase in reflex duration that defines sensitization was statistically identified in two ways:

First, we performed paired t-tests for comparison of each individual post-test reflex (5, 15, 30 and 60 min post attack) to the average of its

two pre-tests, as well as two-sample t-tests to compare experimental post-tests to control (no attack) post-tests. To compensate for the fact that we tested significance of four different post-tests, we adjusted the relevant p-values of each test with a Bonferroni correction.

Second, we used mixed linear models (lme4; Bates, Mächler, Bolker, & Walker, 2014) to test for overall sensitization. To this end, we employed a within-group (Pre/Post, attacked subjects only), as well as a between-group test (Pre/Post $\times$ Attack/NoAttack). Employing these mixed linear models allowed us to compare overall responses among multiple groups as well as assess trends across time. The inclusion of the Bonferroni-corrected t-tests is for convenience, and connecting the statistical tests to the Figs. 1 and 2.

In addition to assessing short-term sensitization trends across time, we also used mixed linear models (on the entire data set, and separately on the four different attack experiments) to measure heterogeneity of sensitization across other categories:

1. Is overall sensitization greater after artificial attack than after natural attack (three-way interaction, Pre/Post $\times$ Attack/NoAttack $\times$ Natural/ArtificialAttack)?
2. Does head attack cause more sensitization than tail attack (three-way interaction Pre/Post $\times$ Attack/NoAttack $\times$ Head/TailAttack)?
3. Is head withdrawal more easily sensitized than tail-mantle withdrawal (three-way interaction: Pre/Post $\times$ Attack/NoAttack $\times$ Head/Tail-mantleWithdrawal)?
4. Site-specific sensitization: Is reflex withdrawal enhanced to stimuli at the anatomical point of attack (head or tail), compared to responses to stimuli at the other end of the test subject (4-way interaction: Pre/Post $\times$ Attack/NoAttack $\times$ Head/TailAttack $\times$ Head/Tail-mantleWithdrawal)?

5. Does site-specific sensitization have an earlier time of onset than general sensitization (3-way interaction: Head/TailAttack $\times$ Head/Tail-mantleWithdrawal $\times$ Time, restricted to post-test responses)?

2.4.2. Long-term sensitization

For each subject, we averaged the two pretest responses and three post-test responses. We used these averages to test for long-term sensitization with three approaches; 1) paired t-tests of the change in reflex duration of attacked animals only (pre- to post-test reflexes), 2) two-sample t-tests of post-test reflexes of attacked vs. not-attacked subjects, 3) two sample t-tests (attacked vs. not attacked) of the average change (average post-test minus the average pre-test) in reflex withdrawal.

2.4.3. Ball-up response

We examined ball-up response durations to successive *Navanax* attacks during the long-term sensitization studies with a repeated measures ANOVA.

3. Results

We confirmed (Paine, 1965; Susswein et al., 1984, 1987; Pennings, 1990) that, unlike the tearing/pinching/poking action of attack by lobster (see 1:52 of Video 1), or fish (Video 2), the attacking cephalaspidean sea slug, *Navanax inermis*, attempts to swallow its molluscan prey whole (Video 3). However, if the latter is too large to swallow (see :59 of Video 1), individual *Navanax* commonly release their *Aplysia* prey some seconds to minutes after attack. By observing head and tail-mantle withdrawal before and after attack, and by directing the attack to the head or tail of subject *Aplysia* (Video 4), we were able to distinguish site-

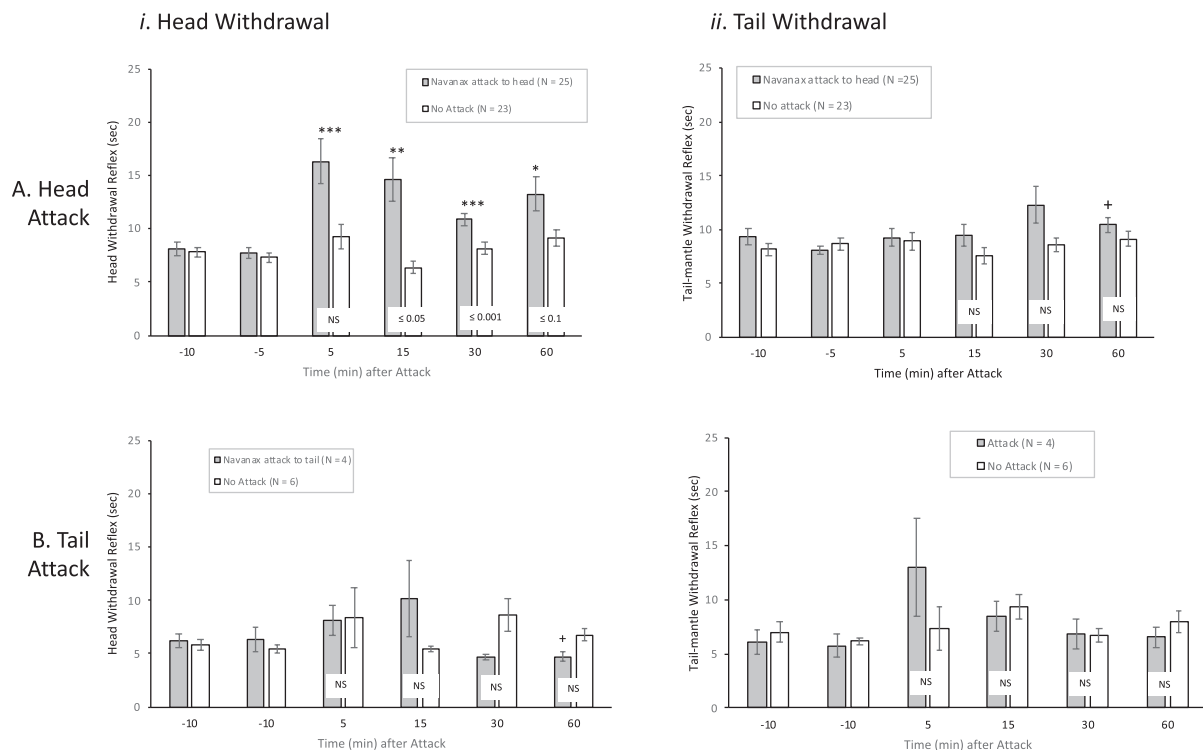


Fig. 1. Sensitization caused by natural *Navanax* attack depended on the anatomical site of the attack (A, B), as well as the site of the test stimulus (i, ii). A. Natural *Navanax* attack to the head of individual *Aplysia* caused immediate pronounced sensitization of head withdrawal (A-i), but only modest sensitization in tail-mantle withdrawal (A-ii). B. Tail attack caused no significant sensitization of either head (B-i) or tail-mantle withdrawal (B-ii). Note that the + symbol above the 60-min test in B-i refers to trending inhibition (not the expected sensitization). Shown are mean reflex durations ($\pm$ standard error of mean), before (−10, −5 min) and after (5, 15, 30, and 60 min) attack by *Navanax* (grey bars, attacked individuals; white bars not-attacked). The symbols above bars refer to p-values from Bonferroni-corrected repeated-measures t-tests of each post-attack withdrawal compared to that subject's average pre-attack withdrawal ($+ < 0.1$, $* < 0.05$, $** < 0.01$, $*** < 0.005$). The probabilities written across bars refer to between-group (*Navanax* attacked vs No-attack) comparisons of post-attack withdrawal with two-sample t-tests.

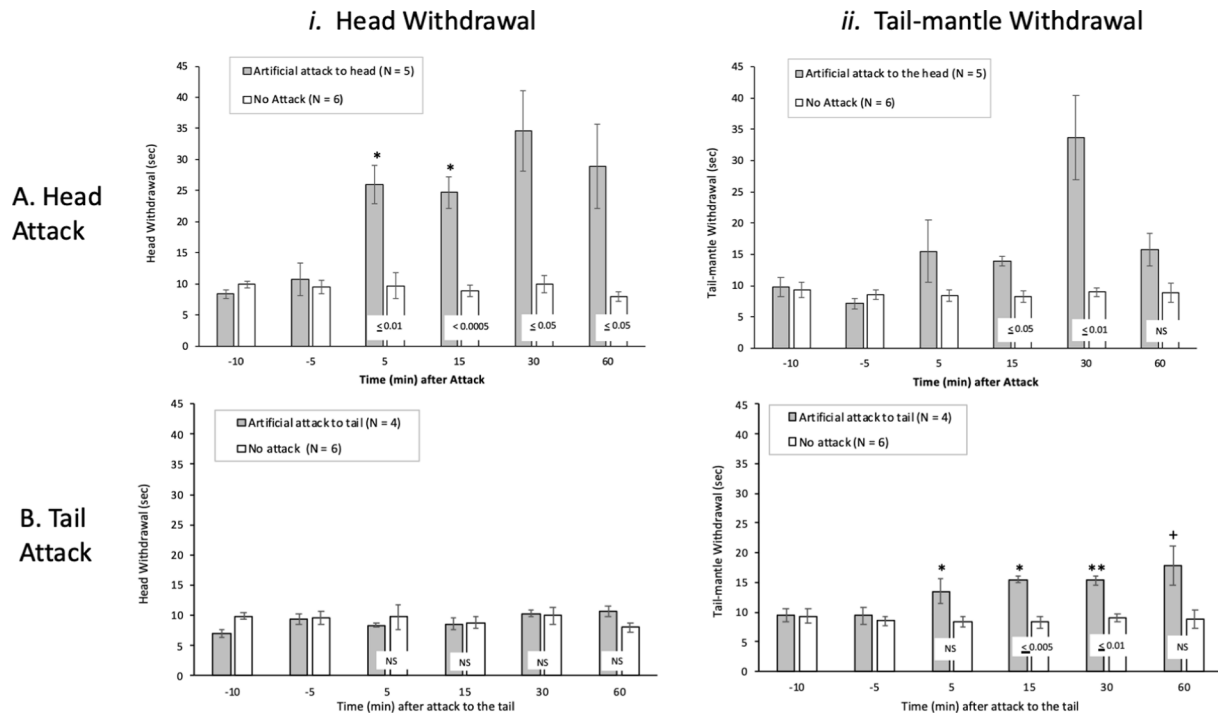


Fig. 2. Artificial *Navanax* attacks produced stronger but qualitatively similar patterns of sensitization as produced by natural *Navanax* attacks (Fig. 1): A. Head withdrawal (i.) was strongly and immediately sensitized by artificial head attack, while tail-mantle withdrawal (ii) was more weakly sensitized. B. Head withdrawal was scarcely affected by tail attack (i.), while tail-mantle withdrawal showed significant immediate sensitization (ii.). The symbols above bars refer to p-values from Bonferroni-corrected repeated-measures t-tests of each post-attack withdrawal compared to that subject's average pre-attack withdrawal (+ < 0.1, * < 0.05, ** < 0.01, *** < 0.005). The probabilities written across bars refers to between-group comparisons with two-sample t-tests. Note change in scale of vertical axis from Fig. 1.

specific (e.g. head withdrawal after head attack) from general sensitization (e.g., tail withdrawal after head attack).

3.1. Short term sensitization

1-A. Natural head attack (Fig. 1A). i. Head withdrawal reflex. Natural *Navanax* attack to the head of subject *Aplysia* caused significant sensitization of head-withdrawal reflexes (Bonferroni-corrected $P < 0.05$: 3 of 4 repeated measures t-tests, 3 of 4 two sample t-tests; Fig. 1A-i). Both mixed linear models (Repeated measures test, Pre/Post, attacked subjects only; and between group test, Pre/Post $\times$ Attack/NoAttack) found a highly significant effect of head attack on head withdrawal (Table 1-1A-i). Comparison of the averaged data (average of two pre-tests, average of 4 post-tests, Fig. S1-A-i) indicated strong sensitization of both reflexes after natural head attack.

ii. Tail-mantle withdrawal reflex. Head attack had a more modest sensitizing effect on tail-mantle withdrawal. Although the averaged data showed significant within-group sensitization (Fig. S1-A-ii, double asterisk, $P = 0.0051$), as well as trending between-group sensitization (Fig. S1-A-ii, $P = 0.066$), reflex withdrawal, tested at each individual time point (after Bonferroni corrections), showed only a trend toward repeated measures sensitization for the last post-test (Fig. 1A-ii, 60 min post, dark bar). In partial agreement with the averaged data, the repeated measures mixed linear model found a significant overall increase in tail-mantle withdrawal after attack (Table 1-1A-ii, $P = 0.018$), but no between-group significance (Table 1-1A-ii).

After these natural head attacks, head withdrawal (average across all post-tests) was proportionally more sensitized ($1.84 \times$ pre-attack) by head attack than was tail-mantle withdrawal ($1.26 \times$ pre-attack; repeated measures t-test, $P = 0.007$), consistent with the hypothesis of site-specific sensitization. The linear mixed model supported this conclusion with a significant three-way interaction (Attack/NoAttack $\times$ Pre/Post $\times$ Head/Tail-mantleWithdrawal; Table 2-1A, $P = 0.0173$).

Table 1

Sensitization caused by natural and artificial attacks to the head and tail of subject *Aplysia* analyzed by mixed linear models (R-project). Shown for each of the four attack groups (natural head attack, natural tail attack, artificial head attack, artificial tail attack) are t-values (sample size) and probabilities for repeated measure analysis (overall comparison of Post vs. Pre in attacked animals only) and between-group (Pre/Post $\times$ Attack/NoAttack) models.

SENSITIZATION				
1. NATURAL ATTACK	A. Head Attack	Repeated Measures	i. Head Withdrawal t(129) = 5.78 P < 0.000001	ii. Tail-mantle Withdrawal t(124) = 2.40. P = 0.0178
		Between Group	t(250) = 4.03 P = 0.00008	t(241) = 1.137 NS
	B. Tail Attack	Repeated Measures	t(22) = .740, NS	t(22) = 1.43 NS
		Between Group	t(46) = -0.193, NS	t(45) = 1.029 NS
2. ARTIFICIAL ATTACK	A. Head Attack	Repeated Measures	t(30) = 5.425, P = 0.000007	t(26) = 2.583 P = 0.0158
		Between Group	t(62) = 3.075, P = 0.003129	t(62) = 3.075 P = 0.0031
	B. Tail Attack	Repeated Measures	t(23) = 1.251 NS	t(24) = 4.799 P = 0.00007
		Between Group	t(59) = 1.121 NS	t(60) = 5.026 P = 0.000005

Table 2

Mixed linear model tests of site-specific sensitization (see Methods); separated into the four different attack experiments (top to bottom): natural head attack, natural tail attack, artificial head attack, and artificial tail attack.

1. Natural Attack	A. Head Attack (sensitizes head more than tail)	$t(491) = 2.39$ P = 0.0173
	B. Tail Attack (sensitizes tail more than head)	NS
2. Artificial Attack	A. Head Attack (sensitizes head more than tail)	$t(128) = 2.031$ P = 0.04433
	B. Tail Attack (sensitizes tail more than head)	$t(119) = 2.4411$ P = 0.016115

1-B. Natural tail attack (Fig. 1B, Table 1-1). *i.* Head withdrawal. Unlike the sensitizing effect of head attack, natural *Navanax* attack to the tail caused no significant increase in the head withdrawal reflex (Fig. 1B-i, Table 1-1B-i). The only post-test that even trended from the pre-test average was the 60 min post-test (Fig. 1B-i), which was slightly ($P < 0.10$) **inhibited** (See also grouped data, Fig. S-1-B-i).

ii. Tail-mantle withdrawal. No significant change in tail-mantle withdrawal was observed after natural tail attack (Fig. 1B-ii; Table 1-1B-ii; Fig. S-1-B-ii).

2-A. Artificial head attack (Fig. 2A, Table 1-2A). *i.* Head withdrawal. Immediately following artificial head attack, head withdrawal was strongly sensitized, with significant paired t-tests on the first two post-tests and significant two-sample t-tests on all four (Fig. 2A-i). Sensitization was also indicated by the linear mixed models (Table 1-2A-i; Repeated measures, Pre/Post only, $P < 0.0001$; Between group, Pre/Post $\times$ Attack/NoAttack, $P = 0.0031$).

ii. Tail-mantle withdrawal. The same artificial head attack that sensitized head withdrawal also sensitized tail-mantle withdrawal (general sensitization), although not as strongly. Although repeated measures t-tests were not significant, two-sample t-tests of reflex withdrawals of attacked sea hares vs non-attacked sea hares were significant at 15 and 30 min post (Fig. 2A-ii), suggesting delayed sensitization. The linear mixed model also indicated significant sensitization of tail-mantle withdrawal after artificial head attack (Table 1-2A-ii; repeated measures, $P = 0.0158$; between-group, $P = 0.0031$).

As with natural head-attack, artificial head-attack sensitized head withdrawal more strongly than it did tail-mantle withdrawal (general linear model, Pre/Post $\times$ Attack/NoAttack $\times$ Head/TailMantleWithdrawal; Table 2-2A, $P = 0.0443$), consistent with site-specific sensitization. The grouped data supported these conclusions as well (Fig. S-1-C): Head withdrawal averaged 1.68 times pre-test, tail-mantle withdrawal 1.22 times pretest, with all t-tests significant.

2-B. Artificial tail attack (Fig. 2B, Table 1-2B). *i.* Head withdrawal. Similar to natural tail-attack (Fig. 1B-i) artificial tail-attack of subject *Aplysia* produced no sensitization in head withdrawal (Fig. 2B-i, Table 1-2B-i).

ii. Tail-mantle withdrawal. Unlike natural tail attack (Fig. 1B-ii), artificial tail attack caused immediate sensitization of tail-mantle withdrawal (Fig. 2B-ii), with three of four post-tests significantly greater than the average pre-test, and two post-tests greater than their respective controls. The mixed linear models also indicated sensitization of tail-mantle withdrawal (Table 1-2B-ii; repeated measures, Pre/Post, Attack only, $P < 0.0001$; between group, Attack/NoAttack $\times$ Pre/Post, $P < 0.0001$). The linear mixed model supported the site-specific sensitization hypothesis (Table 2-2B, Pre/Post $\times$ Attack/NoAttack $\times$ Head/Tail-mantleWithdrawal, $P = 0.0161$), and the grouped data (Fig. S-1-B) clearly demonstrated strong sensitization of tail-mantle withdrawal (1.68 times pre-attack, repeated measures $P = 0.003$, between group $P = 0.002$), with no detectable sensitization of head withdrawal (1.13

times pre attack, repeated measures and between group t-tests non-significant).

3.) Interactions.

Consideration of all four experiments (Figs. 1 and 2) with mixed linear models, demonstrated that:

- sensitization after artificial attack was greater than after natural attack (Pre/Post, attacked only: $t_{400} = 8.133$, $P < 0.0001$; Pre/Post $\times$ Attack/NoAttack: $t_{829} = 6.879$, $P < 0.0001$).
- head attack was overall a slightly more salient sensitizer than was tail attack (attacked subjects only; Pre/Post $\times$ Head/TailAttack; $t_{400} = 1.984$, $P = 0.0479$). This effect was driven primarily by the artificial-attack data ($t_{247} = 3.95$, $P = 0.0001$); it was not strongly supported by natural attack ($t_{297} = 1.088$, NS).
- overall, head withdrawal was more readily sensitized than was tail-mantle withdrawal (three-way interaction: Pre/Post $\times$ Attack/NoAttack $\times$ Head/Tail-mantleWithdrawal, $t_{829} = 2.253$, $P = 0.0245$); this increased sensitivity of head withdrawal to attack was primarily driven by natural attack ($t_{582} = 1.981$, $P = 0.048$); artificial attacks did not sensitize head withdrawal significantly more than they did tail withdrawal ($t_{247} = 1.22$, NS).
- overall, site-specific sensitization (reflex responses to test stimuli delivered to the anatomical site of the eventual attack becoming more sensitized than responses to test stimuli delivered away from the attack site; 4-way interaction: Pre/Post $\times$ Attack/NoAttack $\times$ Head/TailAttack $\times$ Head/Tail-mantleWithdrawal, $t_{829} = 2.853$, $P = 0.0044$) was observed after three of the four kinds of attack (Table 2). There was little evidence that site-specific sensitization was any different between natural vs. artificial attack (5-way interaction: Pre/Post $\times$ Attack/NoAttack $\times$ Head/TailAttack $\times$ Head/Tail-mantleWithdrawal $\times$ Natural/ArtificialAttack, $t_{829} = 1.415$, NS).
- The hypothesis that general sensitization is delayed relative to site-specific sensitization was supported by the data from natural head attack (Fig. 1A-i vs 1A-ii). In particular, the linear mixed model (attack-only, post-only data) confirmed a modest differential timing of sensitization of head (site-specific) vs tail-mantle (general) sensitization (Head/Tail-mantleWithdrawal $\times$ Time, $P = 0.049$). This suggests that site-specific sensitization seen in head withdrawal after head attack (Fig. 1A-i) has a more immediate onset than general sensitization seen in tail-mantle withdrawal (Fig. 1A-ii) following the same attack. Although the mixed linear model analysis of the differential timing hypothesis received little support from the artificial head-attack experiments (Fig. 2A-i vs 2A-ii, NS), the combined (natural and artificial head-attack) data showed a trending differential timing ($P = 0.09$).

3.2. Long-term sensitization

Paradoxical long-term sensitization after multiple attacks. To test a key feature of learning and memory that multiple training experiences produce memory that persists over 24 h (Pinsker, Henning, Carew, & Kandel, 1973; Frost, Castellucci, Hawkins, & Kandel, 1985; Wright, McCance, & Carew, 1996; Kandel, 2012; Mason et al., 2014), we subjected *Aplysia* to four temporally spaced natural *Navanax* attacks to the head (see Methods). Unlike the clear short-term sensitization observed in head withdrawal after a single natural head attack (Fig. 1A-i), there was little evidence of sensitization in head withdrawal 24 h after multiple natural head attacks (Fig. 3-i). Surprisingly, in the same animals, tail-mantle withdrawal, which was only barely sensitized in the short term (Fig. 1A-ii), was significantly sensitized 24 h after multiple attacks to the head (Fig. 3-ii; paired t-test of post vs. pre-tests, $P = 0.0012$). In addition, the average difference score (Post-Pre, Fig. 3-ii) of tail-mantle withdrawal of animals subjected to four natural head attacks was significantly longer than that of control animals (two sample t-test, $P = 0.005$, Fig. 3-ii).

5.) Post-attack “ball-up” duration increased across multiple

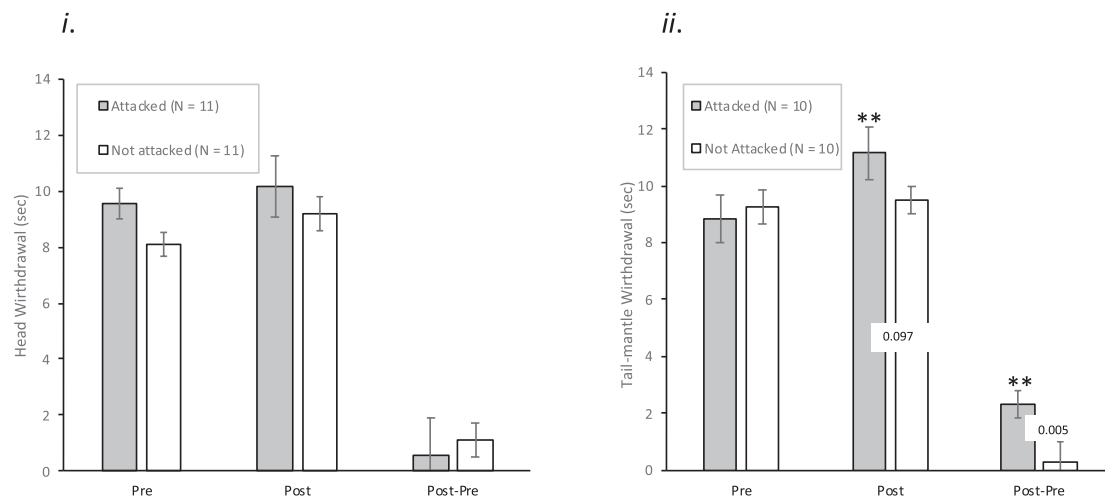


Fig. 3. Paradoxical long-term sensitization after multiple *Navanax* attacks. 24 h after four natural *Navanax* attacks to the head, subject *Aplysia* showed no sensitization in head withdrawal (i). By contrast, modest long-term sensitization was observed in the tail-mantle withdrawal reflex of the same animals (ii). Each individual's "Pre" score was the average of the withdrawals in response to the two pre-tests prior to attack; "Post" indicates the average of three withdrawals in response to the post-tests > 24 h after the last attack. "Post-Pre" indicates the average of each animal's average Post minus its average Pre.

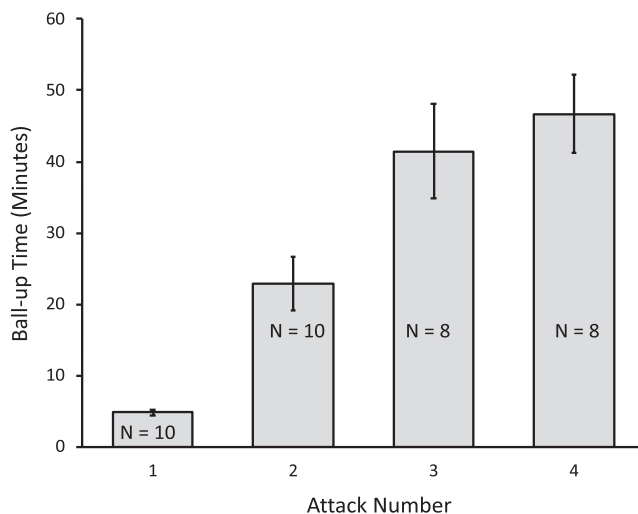


Fig. 4. Duration of the "ball-up" response in attacked (head attacks) subject *Aplysia* increased after each successive attack.

Navanax attacks. During the course of the long-term sensitization training (previous paragraph), the length of time subject *Aplysia* remained in a contracted state following each *Navanax* attack to the head, increased steadily across successive attacks, from about 5 min after the first attack, to over 45 min after the last attack (Fig. 4; repeated measures ANOVA, $F_{1,34} = 55$, $P < 0.0001$). This "wind-up" (Walters, 1994) of the response to successive *Navanax* attacks does not formally represent sensitization; the behavioral withdrawal is not a response to an "otherwise innocuous stimulus," and the attack intervals (<18 h) were less than the minimum 24 h commonly used as a definition of long-term memory (Bailey & Chen, 1983; Frost et al., 1985; Kandel, 2012); it nevertheless supports the idea that long-lasting memory may be generated by repeated *Navanax* head attack.

4. Discussion

Despite the power and utility of *Aplysia* as a model species for the study of mechanisms of learning and memory (Kandel, 2001, 2012), our understanding of the natural adaptive forces shaping such learning and memory in *Aplysia* is meager at best: No naturally occurring noxious

stimulus was known to cause sensitization until Watkins et al. (2010) showed that sub-lethal attack by the California spiny lobster (*Panulirus interruptus*) causes short-term sensitization. We here establish an additional natural sensitization stimulus, attack by the sea slug *Navanax inermis*, and compare these two very different predator-induced behavior changes from both mechanistic and adaptive/evolutionary perspectives.

4.1. Mechanistic perspective

Consideration of *Navanax* attack and its consequences distinguish this noxious stimulus from that used by previous studies. First, in contrast to the effects of both lobster attack (Watkins et al., 2010) and electric shock (Walters, 1987a), *Navanax* attack does not cause lesions in the skin of attacked *Aplysia*. This suggests that activation of surface mechano-sensory nociceptors, observed in previous studies of sensitization using electric shock (see e.g., Illich & Walters, 1997; Walters, 1987a,b, 1991), is unlikely during *Navanax* attack. Second, while copious release of ink and opaline follows both lobster attack (see Video 5; Nusenbaum & Derby, 2010; Love-Chezem et al., 2013) and electric shock (Shapiro, Koester, & Byrne, 1979), these chemical defenses are almost never deployed in response to *Navanax* attack. Third, unlike lobster attack which involves probing and biting its *Aplysia* prey over the entire body, *Navanax* attack is directed onto the point of initial contact (usually head or tail). Finally, although attack by both lobster and *Navanax* causes *Aplysia* to strongly withdraw its head and tail into a compact ball, this "balling up" likely serves different functions. For lobster attack it reduces the severity of bites to the relatively vulnerable head region; for *Navanax* attack it maximizes the prey's cross-sectional area, making it more difficult to swallow whole. How do these differences between *Navanax* and lobster attack contribute to the differing sensitization phenotypes documented in the present study?

4.2. Site-specific vs. general sensitization

Previous experiments in *Aplysia* using electric shock established a dichotomy between **general** sensitization of reflexes to stimuli delivered away from the anatomical site of the shock vs **site-specific** sensitization of reflexes initiated at or near to the site of the shock. Site-specific sensitization (Walters, 1991) causes greater enhancement of reflex withdrawal than does general sensitization (Walters, 1987a,b, 1991). It also tends to be expressed very soon after the noxious stimulus (Walters,

1987b, E.T. Walters pers. comm. to WGW), while the weaker general sensitization is delayed until 20 or 30 min later (Mackey et al., 1987; Marcus et al., 1988; Wright et al., 1992). These differences appear to hold across multiple reflexes and noxious stimuli (Walters, 1991).

We note that testing for site-specific sensitization after lobster attack (Watkins et al., 2010; Mason et al., 2014) is impossible, inasmuch as lobster attack distributes noxious stimuli over the entire surface of the *Aplysia* body. By contrast, because *Navanax* attack is localized to either head or tail, we were here able to demonstrate the first known instance of site-specific sensitization caused by a natural stimulus: Head-attack induced sensitization of head withdrawal that was significantly greater than sensitization of tail withdrawal, and expressed much more quickly after the attack (5 min post-attack, Fig. 1Ai, Fig. 2Ai) than observed by previous investigators of general sensitization (Marcus et al., 1988, 20 min; Wright et al., 1992, 30 min; Watkins et al., 2010, 30 min).

The same natural head attack caused weaker but significant general sensitization in the tail-mantle withdrawal reflex (Fig. 1A-ii). This weaker general sensitization also showed modest evidence of delay, relative to sensitization of head withdrawal after the same head attack (Fig. 1A-ii, $P = 0.049$), consistent with previous studies of general sensitization (see previous paragraph).

What might be the neural basis of these two kinds of sensitization (strong, immediate site-specific sensitization; weaker, delayed general sensitization)? For site-specific sensitization, Walters (1987a,b, 1991) reframed previous work on mechanisms of classical conditioning (Hawkins, Abrams, Carew, & Kandel, 1983; Walters, Carew, & Kandel, 1981) to propose that simultaneous activity of sensory and modulatory interneurons activated by local strong electric shock produces stronger and more immediate heterosynaptic facilitation of sensory-neuron synapses leading to more robust and immediate behavioral sensitization near the site of the noxious stimulus. This activity dependent enhancement of facilitation is likely supported by changes in both sensory (Walters, 1987a,b, 1991; Kandel, 2012) and motor neurons (Eliot, Hawkins, Kandel, & Schacher, 1994; Murphy & Glanzman, 1997). We propose that similar activity dependent enhanced facilitation is likely contributing to immediate site-specific sensitization observed in three (Figs. 1A-i, 2A-i, and 2B-ii) of our four experiments.

What might be the neural basis of general sensitization? We propose that the weaker, sensitization observed in tail-mantle withdrawal after natural head attack (Fig. 1A-ii) may be due to suppression of withdrawal-circuit neurons by inhibitory interneurons activated by noxious stimuli (Hawkins, Castellucci, & Kandel, 1981a,b; Wright, Marcus, & Carew, 1991; Buonomano, Cleary, & Byrne, 1992; Fischer & Carew, 1993; Frost & Kandel, 1995; Wright & Carew, 1995; Xu, Cleary, & Byrne, 1994, 1995; Cleary, Lee, & Byrne, 1998; Fischer, Blazis, Priver, & Carew, 1997). Because this suppression is usually short-lived, its waning likely results in the delayed emergence of sensitization (Wright et al., 1991; Wright & Carew, 1995). Longer-lasting increases in motor-neuron sensitivity (Glanzman, 2008) may also contribute to general sensitization at the 30 and 60 min time points (Figs. 1A, 2A).

4.3. Adaptive/ecological/evolutionary perspective

What adaptive hypotheses are suggested by these observations? Because behavioral interactions of *Aplysia* with spiny-lobsters are markedly different from those with *Navanax*, these interactions are worth comparing to better understand their ecological context. The initial response of subject *Aplysia* to lobster attack includes the release of defensive fluids (ink and opaline, see Video 5). Depending on the hunger state of the lobster (Berriman et al., 2015; see also Kicklighter, Shabani, Johnson, & Derby, 2005, 2007; Derby & Aggio, 2011), this can result in the lobster releasing the sea hare (see 0:59 of Video 1) and turning its attention to cleaning or eating the ink/opaline off its antennae and pereopods (Kicklighter et al., 2005; Derby, 2007). Lobsters may be thus diverted from attack for several to many minutes (C. Derby, pers. comm.

to WGW). During this time, attacked sea hares often “gallop” (Jahan-Parwar & Fredman, 1979; Leonard & Lukowiak, 1984, 1986; Walters, 1991) as fast as possible from the site of the attack (Wachtel & Impelman, 1973; Kicklighter, Kamio, Nguyen, Germann, & Derby, 2011; Derby & Aggio, 2011, W. G. Wright, pers. obs.), thereby reducing the probability of the lobster relocating them after it has recovered from the ink/opaline. This post-attack escape behavior by *Aplysia* reasonably requires full participation of the head and tail to realize maximal speed (Jahan-Parwar & Fredman, 1979; Leonard & Lukowiak, 1984, 1986); protracted withdrawal of either is likely to slow the escape. After sufficient distance has been attained by the galloping sea hare, sensitization is adaptively activated to a wide range of stimuli to help protect the once-attacked sea hare from further attacks (Walters, 1991). Thus, chemical deterrence, coordinated with escape behavior, seems a reasonable explanation for the weaker/delayed general sensitization observed in *Aplysia* after lobster attack.

In the case of *Navanax* head-attack (Fig. 1A-ii), similar considerations as after lobster attack are somewhat awkward, inasmuch as *Navanax* attack does not cause release of ink, and thus no distraction of the attacking *Navanax*. Although we noticed a “refractory period” in the propensity of *Navanax* to attack a second time, we did not quantify or describe it. We also note that, unlike lobsters, whose mobility is orders of magnitude more rapid than that of sea hare, the mobility of a *Navanax* predator is roughly equivalent to that of its *Aplysia* prey; thus, whereas post-attack galloping can reduce the chances of a lobster predator finding its prey once it has recovered from the chemical defenses, a similar rapid locomotory response by the sea hare might actually outrun a pursuing *Navanax* predator. The possible role of escape locomotion after *Navanax* attack receives some support from ancillary observations of *Aplysia* behavior following natural *Navanax* attack to the tail of subject *Aplysia* (which caused no sensitization of either reflex). First, water-jet stimuli to the tail caused once-attacked *Aplysia* to move rapidly away. Second, tail attack caused a statistically trending **decrease** in the duration of head withdrawal to water-jet stimuli 60 min after the attack (Fig. 1B-i), consistent with ‘flight readiness’. We hasten to add that these are only small pieces to the overall puzzle of the ecology of *Navanax*-*Aplysia* interactions.

What, then, is the adaptive value of the sensitization observed in head withdrawal after head attack? We favor the idea that sensitized head withdrawal after *Navanax* attack may protect the subject *Aplysia* by maximizing its cross-sectional area, and thus its chances for being too large for a particular *Navanax* to ingest (Pennings, 1990). Perhaps equally important, stronger head withdrawal after head-attack likely protects the already challenged internal tissues of the recently attacked *Aplysia* from further injury.

4.4. Unexpected sensitization results

The above observations of short-term sensitization after *Navanax* attack are roughly consistent with previous research on sensitization using electric shock as a noxious stimulus (initial contraction in response to attack, strong immediate site-specific sensitization, weaker delayed general sensitization). However, our experiments also produced three previously unreported results.

First, the observation that attack to the head produces more overall (head and tail withdrawal combined) sensitization than does tail attack (compare overall Figs. 1A vs. 1B, and 2A vs. 2B) is likely due to the presence of hard parts, such as the radular and gizzard teeth (Howells, 1942) in the head region. This may damage internal tissue during the *Navanax* squeeze, thereby activating or damaging internal nociceptor neurons (Walters, pers. comm. to WGW). By contrast, the tissue of the *Aplysia* tail is relatively homogeneous, with no sharp or vulnerable internal organs, likely rendering this region less vulnerable to *Navanax* attack.

Second, head withdrawal showed more short-term sensitization (across all attack experiments, Figs. 1 and 2) than did tail-mantle

withdrawal. Although we are presently ignorant of a cellular mechanism for this overall anatomical heterogeneity of sensitization after *Navanax* attack, this finding makes sense from an adaptive perspective; the head region carries sensitive functional components (the central nervous system, buccal mass, gizzard, etc.), whose protection from future attacks is likely to be particularly critical.

Third, and most surprising, we found that four spaced *Navanax* head-attacks, any one of which can cause robust short-term sensitization of the head-withdrawal reflex (Fig. 1A-i), failed to cause long-term sensitization of head withdrawal (sensitization lasting more than 24 h, Fig. 3-i). This is in sharp contrast to similarly spaced bouts of electric shock (Pinsker et al., 1973; Walters, 1987a; Kandel, 2001) or lobster attack (Mason et al., 2014), both of which produce robust long-term sensitization. The absence of long-term sensitization of head withdrawal after multiple head attacks by *Navanax* becomes even more perplexing in light of the fact that these same head attacks actually caused significant long-term sensitization in tail-mantle withdrawal (Fig. 3-ii). Thus, the only long-term sensitization observed after multiple *Navanax* attacks was restricted to an anatomical site (tail) more distal from the site of attack (head). These long-term sensitization findings contrast most specifically with research by Walters (Walters, 1987a, 1991), which showed the exact opposite: enhanced long-term site-specific sensitization persists much longer than does general long-term sensitization. This makes questions of neural mechanisms of long-term sensitization particularly interesting: Many of the short-term cellular changes in the neurons of reflex circuits supporting short-term sensitization, e.g., serotonin induced increase in transmitter release (Kandel, 2001) and post-synaptic increases in the motor-neuron response to that transmitter release (Glanzman, 2008), are in series with, and in some cases thought to be necessary preconditions of, mechanisms underlying long-term sensitization (e.g., growth of synaptic processes between sensory and motor neurons; Glanzman, Kandel, & Schacher, 1990; Kandel, 2012; K. Martin, 2002; K.C. Martin et al., 1997).

How might multiple *Navanax* attacks to the head, each of which can so effectively causes short-term sensitization in head withdrawal, fail to produce long-term sensitization in the same reflex? Perhaps the intense squeeze of the head by *Navanax* attack alters the blood delivery of serotonin or other critical neuromodulator to relevant synapses in the head-ganglia (buccal, cerebral, pleural and pedal), possibly preventing long-term sensitization of head withdrawal, without impacting either short-term sensitization, which is primarily effected by locally released serotonin (Wright, Jones, Sharp, & Maynard, 1995; Marinesco & Carew, 2002), or long-term sensitization in tail-mantle withdrawal (because the tail wasn't squeezed).

To conclude, these experiments and observations document short and long-term sensitization in *Aplysia*, caused by *Navanax* attack, adding such attack to a short list of natural stimuli that can cause sensitization (present study; Watkins et al., 2010). They also document, for the first time, a natural stimulus that causes site-specific sensitization (enhanced sensitization of reflexes at the site of the attack, Walters, 1991). Finally, the failure of four spaced head-attacks to cause long-term sensitization of head withdrawal (Fig. 3-i), when any one of them can cause robust short-term sensitization is puzzling, particularly in view of the fact that the same head attacks succeed in producing long-term sensitization of the tail reflex at the opposite (tail) end of subject *Aplysia* (Fig. 3-ii).

Together, these observations support the importance of ecological context in providing a complete understanding of learning and memory in a model species. We maintain that previous laboratory experiments using artificial noxious stimuli are better placed into an adaptive, and even mechanistic, context by explorations of parallel effects using natural stimuli.

CRediT authorship contribution statement

Chrissy Pepino: Investigation, Conceptualization, Methodology, Validation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Supervision, Visualization. **Cyril Rakovski:** Investigation, Methodology, Software, Validation, Formal analysis. **Candace Gutierrez:** Investigation, Methodology, Data curation. **Amanda Rodriguez:** Investigation, Methodology, Data curation. **Scott Tillett:** Investigation. **Cassandra Berriman:** Investigation. **Maria Mason:** Investigation, Methodology, Formal analysis, Writing – review & editing. **Amanda Watkins Ingalls:** Investigation, Writing – review & editing. **Rebecca Emshwiler:** Investigation, Methodology. **Sarah Christian Scher:** Investigation, Methodology. **Virginia Zachary:** Investigation. **Lucy Lee:** Investigation. **Sarah Johnson:** Investigation, Methodology. **Kendra Olsen:** Investigation, Methodology. **William G. Wright:** Investigation, Conceptualization, Methodology, Software, Validation, Formal analysis, Resources, Data curation, Writing – original draft, Writing – review & editing, Supervision, Visualization, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The data for these findings were gathered over the course of nearly a decade in the Marine Invertebrate Behavior lab at Chapman University. We thank Janet Leonard, Steven Pennings, Elizabeth Wilder, Frances Ross, and Jaclyn Greenwood for their helpful comments, and an anonymous reviewer for suggesting the squeeze hypothesis to explain paradoxical long-term sensitization. We also thank Terry Walters, Emilie Marcus, and Deborah Bird, for critical readings of earlier versions, as well as generous discussions and email exchanges. This work was supported by the National Science Foundation (IBN-0131743, IOS-0721800).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.nlm.2021.107542>.

References

- Antzoulatos, E. G., & Byrne, J. H. (2004). Learning insights transmitted by glutamate. *Trends in Neurosciences*, 27(9), 555–560.
- Bailey, C. H., & Chen, M. (1983). Morphological basis of long-term habituation and sensitization in *Aplysia*. *Science*, 220(4592), 91–93. <https://doi.org/10.1126/science.6828885>
- Barco, A., Bailey, C. H., & Kandel, E. R. (2006). Common molecular mechanisms in explicit and implicit memory. *Journal of Neurochemistry*, 97(6), 1520–1533. <https://doi.org/10.1111/j.1471-4159.2006.03870>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2014). Fitting linear mixed-effects models using lme4. *arXiv preprint arXiv:1406.5823*.
- Berriman, J. S., Kay, M. C., Reed, D. C., Rassweiler, A., Goldstein, D. A., & Wright, W. G. (2015). Shifts in attack behavior of an important kelp forest predator within marine reserves. *Marine Ecology Progress Series*, 522, 193–201.
- Blair, G. M., & Seapy, R. R. (1972). Selective predation and prey location in the sea slug *Navanax inermis*. *Veliger*, 15(2), 119–124.
- Buonomano, D. V., Cleary, L. J., & Byrne, J. H. (1992). Inhibitory neuron produces heterosynaptic inhibition of the sensory-to-motor neuron synapse in *Aplysia*. *Brain Research*, 577(1), 147–150.
- Cleary, L. J., Lee, W. L., & Byrne, J. H. (1998). Cellular correlates of long-term sensitization in *Aplysia*. *Journal of Neuroscience*, 18(15), 5988–5998. Retrieved from <Go to ISI>://WOS:000074971800038.

- Derby, C. D. (2007). Escape by inking and secreting: Marine molluscs avoid predators through a rich array of chemicals and mechanisms. *Biological Bulletin*, 213(3), 274–289. Retrieved from <Go to ISI>://000251865600006.
- Derby, C. D., & Aggio, J. F. (2011). The Neuroecology of Chemical Defenses. *Integrative and Comparative Biology*, 51(5), 771–780.
- Eliot, L. S., Hawkins, R. D., Kandel, E. R., & Schacher, S. (1994). Pairing-specific, activity-dependent presynaptic facilitation at *Aplysia* sensory-motor neuron synapses in isolated cell-culture. *Journal of Neuroscience*, 14(1), 368–383. Retrieved from <Go to ISI>://WOS:A1994MQ11800033.
- Fischer, T. M., & Carew, T. J. (1993). Activity-dependent potentiation of recurrent inhibition: A mechanism for dynamic gain control in the siphon withdrawal reflex of *Aplysia*. *Journal of Neuroscience*, 13(3), 1302–1314.
- Fischer, T. M., Blazis, D. E., Privner, N. A., & Carew, T. J. (1997). Metaplasticity at identified inhibitory synapses in *Aplysia*. *Nature*, 389(6653), 860–865.
- Frost, W. N., & Kandel, E. R. (1995). Structure of the network mediating siphon-elicited siphon withdrawal in *Aplysia*. *Journal of Neurophysiology*, 73(6), 2413–2427.
- Frost, W. N., Castellucci, V. F., Hawkins, R. D., & Kandel, E. R. (1985). Monosynaptic connections made by the sensory neurons of the gill-and siphon-withdrawal reflex in *Aplysia* participate in the storage of long-term memory for sensitization. *Proceedings of the National Academy of Sciences*, 82(23), 8266–8269.
- Glanzman, D. L., Kandel, E. R., & Schacher, S. (1990). Target-dependent structural changes accompanying long-term synaptic facilitation in *Aplysia* neurons. *Science*, 249(4970), 799–802.
- Glanzman, D. L. (2008). New tricks for an old slug: The critical role of postsynaptic mechanisms in learning and memory in *Aplysia*. In W. S. Sossin, J. C. Lacaille, V. F. Castellucci, & S. Belleville (Eds.), *Essence of Memory* (Vol. 169, pp. 277–292). Elsevier.
- Hawkins, R. D., Castellucci, V. F., & Kandel, E. R. (1981). Interneurons involved in mediation and modulation of gill-withdrawal reflex in *Aplysia*. I. Identification and characterization. *Journal of Neurophysiology*, 45(2), 304–314.
- Hawkins, R. D., Castellucci, V. F., & Kandel, E. R. (1981). Interneurons involved in mediation and modulation of gill-withdrawal reflex in *Aplysia*. II. Identified neurons produce heterosynaptic facilitation contributing to behavioral sensitization. *Journal of Neurophysiology*, 45(2), 315–326. Retrieved from <Go to ISI>://WOS:A1981LE07500010.
- Hawkins, R. D., Abrams, T. W., Carew, T. J., & Kandel, E. R. (1983). A cellular mechanism of classical conditioning in *Aplysia*: Activity-dependent amplification of presynaptic facilitation. *Science*, 219(4583), 400–405. Retrieved from <Go to ISI>://WOS:A1983PY51600027.
- Howells, H. (1942). The structure and function of the alimentary canal of *Aplysia punctata*. *Journal of Cell Science*, 2(331), 357–397.
- Illich, P. A., & Walters, E. T. (1997). Mechanosensory neurons innervating *Aplysia* siphon encode noxious stimuli and display nociceptive sensitization. *Journal of Neuroscience*, 17(1), 459–469. Retrieved from <Go to ISI>://WOS:A1997WJ67700040.
- Jahan-Parwar, B., & Fredman, S. M. (1979). Neural control of locomotion in *Aplysia*: Role of the central ganglia. *Behavioral and neural biology*, 27(1), 39–58.
- Kamio, M., Grimes, T. V., Hutchins, M. H., van Dam, R., & Derby, C. D. (2010). The purple pigment aplysiocyanin in sea hare ink deters predatory blue crabs through their chemical senses. *Animal Behaviour*, 80(1), 89–100.
- Kandel, E. R. (2001). Neuroscience - The molecular biology of memory storage: A dialogue between genes and synapses. *Science*, 294(5544), 1030–1038.
- Kandel, E. R. (2012). The molecular biology of memory: cAMP, PKA, CRE, CREB-1, CREB-2, and CPEB. *Molecular Brain*, 5.
- Kicklighter, C. E., Shabani, S., Johnson, P. M., & Derby, C. D. (2005). Sea hares use novel antipredatory chemical defenses. *Current Biology*, 15(6), 549–554. <https://doi.org/10.1016/j.cub.2005.01.057>
- Kicklighter, C. E., Germann, M., Kamio, M., & Derby, C. D. (2007). Molecular identification of alarm cues in the defensive secretions of the sea hare *Aplysia californica*. *Animal Behaviour*, 74(5), 1481–1492. <https://doi.org/10.1016/j.anbehav.2007.02.015>
- Kicklighter, C. E., Kamio, M., Nguyen, L., Germann, M. W., & Derby, C. D. (2011). Mycosporine-like amino acids are multifunctional molecules in sea hares and their marine community. *Proceedings of the National Academy of Sciences of the United States of America*, 108(28), 11494–11499.
- Leonard, J. L., & Lukowiak, K. (1984). An ethogram of the sea slug, *Navanax inermis* (Gastropoda, Opisthobranchia). *Zeitschrift Fur Tierpsychologie-Journal of Comparative Ethology*, 65(4), 327–345.
- Leonard, J. L., & Lukowiak, K. (1986). The behavior of *Aplysia californica* Cooper (Gastropoda; Opisthobranchia): I. *Ethogram*. *Behaviour*, 98(1–4), 320–360.
- Love-Cheezem, T., Aggio, J. F., & Derby, C. D. (2013). Defense through sensory inactivation: Sea hare ink reduces sensory and motor responses of spiny lobsters to food odors. *Journal of Experimental Biology*, 216(8), 1364–1372.
- Mackey, S. L., Glanzman, D. L., Small, S. A., Dyke, A. M., Kandel, E. R., & Hawkins, R. D. (1987). Tail shock produces inhibition as well as sensitization of the siphon-withdrawal reflex of *Aplysia*: Possible behavioral role for presynaptic inhibition mediated by the peptide Phe-Met-Arg-Phe-NH₂. *Proceedings of the National Academy of Sciences of the United States of America*, 84(23), 8730–8734. <https://doi.org/10.1073/pnas.84.23.8730>
- Marcus, E., Nolen, T., Rankin, C., & Carew, T. (1988). Behavioral dissociation of dishabituation, sensitization and inhibition in *Aplysia*. *Science*, 241(4862), 210–213.
- Mason, M. J., Watkins, A. J., Wakabayashi, J., Buechler, J., Pepino, C., Brown, M., & Wright, W. G. (2014). Connecting model species to nature: Predator-induced long-term sensitization in *Aplysia californica*. *Learning & Memory*, 21(8), 363–367. <https://doi.org/10.1101/lm.034330.114>
- Marinesco, P., & Carew, T. J. (2002). Serotonin release evoked by tail nerve stimulation in the CNS of *Aplysia*: Characterization and relationship to heterosynaptic plasticity. *Journal of Neuroscience*, 22(6), 2299–2312. Retrieved from <Go to ISI>://WOS:000174379900029.
- Marinesco, S., Duran, K. L., & Wright, W. G. (2003). Evolution of learning in three aplysiid species: Differences in heterosynaptic plasticity contrast with conservation in serotonergic pathways. *Journal of Physiology-London*, 550(1), 241–253. <https://doi.org/10.1113/jphysiol.2003.038356>
- Martin, K. C., Casadio, A., Zhu, H., Yaping, E., Rose, J. C., Chen, M., ... Kandel, E. R. (1997). Synapse-specific, long-form facilitation of *Aplysia* sensory to motor synapses: A function for local protein synthesis in memory storage. *Cell*, 91(7), 927–938. [https://doi.org/10.1016/s0092-8674\(00\)80484-5](https://doi.org/10.1016/s0092-8674(00)80484-5)
- Martin, K. (2002). Synaptic tagging during synapse-specific long-term facilitation of *Aplysia* sensory-motor neurons. *Neurobiology of Learning and Memory*, 78(3), 489–497.
- Murphy, G. G., & Glanzman, D. L. (1997). Mediation of classical conditioning in *Aplysia californica* by long-term potentiation of sensorimotor synapses. *Science*, 278(5337), 467–471.
- Murray, M. J., & Lewis, E. R. (1974). Sensory control of prey capture in *Navanax inermis*. *The Veliger*, 17(2), 156–158.
- Nolen, T. G., Johnson, P. M., Kicklighter, C. E., & Capo, T. (1995). Ink secretion by the marine snail *Aplysia californica* enhances its ability to escape from a natural predator. *Journal of Comparative Physiology a-Sensory Neural and Behavioral Physiology*, 176(2), 239–254. Retrieved from <Go to ISI>://WOS:A1995QE59300011.
- Nusnbaum, M., & Derby, C. D. (2010). Ink secretion protects sea hares by acting on the olfactory and nonolfactory chemical senses of a predatory fish. *Animal Behaviour*, 79(5), 1067–1076. <https://doi.org/10.1016/j.anbehav.2010.01.022>
- Paine, R. T. (1965). Natural history, limiting factors and energetics of the opisthobranch *Navanax inermis*. *Ecology*, 46(5), 603–619.
- Penning, S. C. (1990). Predator-prey interactions in opisthobranch gastropods: Effects of prey body size and habitat complexity. *Marine Ecology-Progress Series*, 62(1–2), 95–101. Retrieved from <Go to ISI>://WOS:A1990CZ63900011.
- Penning, S. C., Nastisch, S., & Paul, V. J. (2001). Vulnerability of sea hares to fish predators: Importance of diet and fish species. *Coral Reefs*, 20(3), 320–324. Retrieved from <Go to ISI>://WOS:000172560800021.
- Pinsker, H. M., Henning, W. A., Carew, T. J., & Kandel, E. R. (1973). Long-term sensitization of a defensive withdrawal reflex in *Aplysia*. *Science*, 182(4116), 1039–1042.
- Rahn, E. J., Guzman-Karlsson, M. C., & Sweatt, D. J. (2013). Cellular, molecular, and epigenetic mechanisms in non-associative conditioning: Implications for pain and memory. *Neurobiology of Learning and Memory*, 105, 133–150.
- Reissner, K. J., Shobe, J. L., & Carew, T. J. (2006). Memory nodes in memory processing: Insights from *Aplysia*. *Cellular and Molecular Life Sciences*, 63(9), 963–974. <https://doi.org/10.1007/s00018-006-6022-x>
- Roberts, A. C., & Glanzman, D. L. (2003). Learning in *Aplysia*: Looking at synaptic plasticity from both sides. *Trends in Neurosciences*, 26(12), 662–670. <https://doi.org/10.1016/j.tins.2003.09.014>
- Rogers, C., De Nys, R., & Steinberg, P. (2000). Predation on juvenile *Aplysia parvula* and other small anaspidan, ascoglossan, and nudibranch gastropods by pycnogonids. *Veliger*, 43(4), 330–337.
- Shapiro, E., Koester, J., & Byrne, J. H. (1979). *Aplysia* ink release: central locus for selective sensitivity to long-duration stimuli. *Journal of Neurophysiology*, 42(5), 1223–1232.
- Susswein, A. J., Achatuv, Y., Cappell, M. S., Spray, D. C., & Bennett, M. V. L. (1984). Pharyngeal movements during feeding sequences in *Navanax inermis*: a cinematographic analysis. *J. comp Physiol. A*, 155, 209–218.
- Susswein, A. J., Achatuv, Y., Cappell, M. S., & Bennett, M. V. L. (1987). Pharyngeal movements during feeding sequences of *Navanax inermis* (Gastropoda: Opisthobranchia) in successive stages of dissection. *Journal of Experimental Biology*, 128, 323–333.
- Sutton, M. A., Bagnall, M. W., Sharma, S. K., Shobe, J., & Carew, T. J. (2004). Intermediate-term memory for site-specific sensitization in *Aplysia* is maintained by persistent activation of protein kinase. *Journal of Neuroscience*, 24(14), 3600–3609. <https://doi.org/10.1523/jneurosci.1134-03.2004>
- Wachtel, H., & Impelman, D. (1973). Galloping escape behavior in *Aplysia californica*. *In Fed. Proc.*, 32(3), 368.
- Watkins, A. J., Goldstein, D. A., Lee, L. C., Pepino, C. J., Tillett, S. L., Ross, F. E., Wilder, E. M., Zachary, V. A., & Wright, W. G. (2010). Lobster attack induces sensitization in the sea hare, *Aplysia californica*. *Journal of Neuroscience*, 30(33), 11028–11031.
- Walters, E. T., Carew, T. J., & Kandel, E. R. (1981). Associative learning in *Aplysia*: Evidence for conditioned fear in an invertebrate. *Science*, 211(4481), 504–506. <https://doi.org/10.1126/science.7192881>
- Walters, E. T. (1987a). Site-specific sensitization of defensive reflexes in *Aplysia*: A simple model of long-term hyperalgesia. *Journal of Neuroscience*, 7(2), 400–407.
- Walters, E. T. (1987b). Multiple sensory neuronal correlates of site-specific sensitization in *Aplysia*. *Journal of Neuroscience*, 7(2), 408–417.
- Walters, E. T. (1991). A functional, cellular, and evolutionary model of nociceptive plasticity in *Aplysia*. *Biological Bulletin*, 180(2), 241–251.
- Walters, E. T. (1994). Injury-related behavior and neuronal plasticity: An evolutionary perspective on sensitization, hyperalgesia, and analgesia. *International review of neurobiology*, 36, 325–427.
- Wright, W. G., Marcus, E. A., & Carew, T. J. (1991). A cellular analysis of inhibition in the siphon withdrawal reflex of *Aplysia*. *Journal of Neuroscience*, 11(8), 2498–2509.
- Wright, W. G., McCance, E. F., Theresa, L., & Carew, T. J. (1992). Delayed-onset sensitization emerges after dishabituation in developing *Aplysia*. *Behavioral and neural biology*, 57(2), 170–174.

- Wright, W. G., Jones, K., Sharp, P., & Maynard, B. (1995). Widespread anatomical projections of the serotonergic modulatory neuron, CB1, in Aplysia. *Invertebrate neuroscience : IN*, *1*(2), 173–183. <https://doi.org/10.1007/BF02331914>
- Wright, W. G., & Carew, T. J. (1995). A single identified interneuron gates tail-shock induced inhibition in the siphon withdrawal reflex of Aplysia. *Journal of Neuroscience*, *15*(1), 790–797.
- Wright, W. G., McCance, E. F., & Carew, T. J. (1996). Developmental emergence of long-term memory for sensitization in Aplysia. *Neurobiology of Learning and Memory*, *65* (3), 261–268. Retrieved from <Go to ISI>://WOS:A1996UH27300009.
- Wright, W. G. (1998). Evolution of nonassociative learning: Behavioral analysis of a phylogenetic lesion. *Neurobiology of Learning and Memory*, *69*(3), 326–337. <https://doi.org/10.1006/nlme.1998.3829>
- Xu, Y., Cleary, L. J., & Byrne, J. H. (1994). Identification and characterization of pleural neurons that inhibit tail sensory neurons and motor neurons in Aplysia: Correlation with FMRFamide immunoreactivity. *Journal of Neuroscience*, *14*(6), 3565–3577.
- Xu, Y., Pieroni, J. P., Cleary, L. J., & Byrne, J. H. (1995). Modulation of an inhibitory interneuron in the neural circuitry for the tail withdrawal reflex of Aplysia. *Journal of Neurophysiology*, *73*(3), 1313–1318.